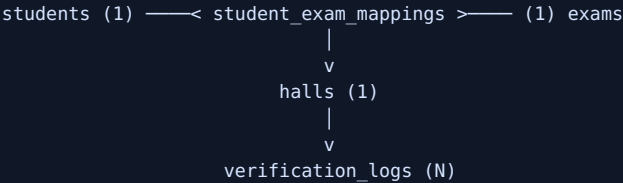


Backend Schema Document

Product: ExamGuard — Smart Exam Allocation & Identity Verification System **Document Version:** 1.0 **Date:** August 14, 2026
Database: PostgreSQL 15+ (relational system of record), Redis 7+ (cache / tokens / pub-sub)

1. Entity-Relationship Overview



One student can have many exam mappings; one exam can have many student mappings; one hall can be referenced by many mappings across many exams; each mapping has zero-or-more verification log entries (normally exactly one terminal entry, occasionally more for flagged conflicts).

2. Table Definitions

2.1 students

```
CREATE TABLE students (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  roll_number VARCHAR(50) UNIQUE NOT NULL,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,
  photo_url TEXT NOT NULL,
  department VARCHAR(100) NOT NULL,
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_students_roll_number ON students(roll_number);
CREATE INDEX idx_students_department ON students(department);
```

Column	Type	Notes
id	UUID	Primary key
roll_number	VARCHAR(50)	Unique login identifier
full_name	VARCHAR(100)	Display name
email	VARCHAR(100)	Unique, used for password reset
password_hash	TEXT	argon2id hash, never returned by API
photo_url	TEXT	Object-storage key; served via signed URL, not raw
department	VARCHAR(100)	Used for roster filtering during mapping

2.2 halls

```
CREATE TABLE halls (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  building_name VARCHAR(100) NOT NULL,
  room_number VARCHAR(20) NOT NULL,
  floor_level INT NOT NULL,
  capacity INT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  UNIQUE(building_name, room_number)
);
```

2.3 exams

```
CREATE TABLE exams (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  course_code VARCHAR(20) NOT NULL,
  course_title VARCHAR(150) NOT NULL,
  exam_date DATE NOT NULL,
  start_time TIME NOT NULL,
  end_time TIME NOT NULL,
  reveal_threshold_minutes INT NOT NULL DEFAULT 30,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_exams_date ON exams(exam_date);
```

2.4 student_exam_mappings

```
CREATE TABLE student_exam_mappings (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  student_id UUID REFERENCES students(id) ON DELETE CASCADE,
  exam_id UUID REFERENCES exams(id) ON DELETE CASCADE,
  hall_id UUID REFERENCES halls(id) ON DELETE CASCADE,
  seat_number VARCHAR(20) NOT NULL,
  barcode_token VARCHAR(255) UNIQUE NOT NULL,
  reveal_unlocked BOOLEAN NOT NULL DEFAULT FALSE,
  used_at TIMESTAMP,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  UNIQUE(student_id, exam_id),
  UNIQUE(exam_id, hall_id, seat_number)
);

CREATE INDEX idx_mappings_exam_hall ON student_exam_mappings(exam_id, hall_id);
CREATE INDEX idx_mappings_student ON student_exam_mappings(student_id);
CREATE INDEX idx_mappings_barcode_token ON student_exam_mappings(barcode_token);
```

The `UNIQUE(exam_id, hall_id, seat_number)` constraint guarantees no two students can ever be double-booked into the same seat for the same exam at the database level, independent of application logic.

2.5 verification_logs

```
CREATE TABLE verification_logs (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  mapping_id UUID REFERENCES student_exam_mappings(id),
  invigilator_id UUID NOT NULL REFERENCES invigilators(id),
  client_event_id UUID NOT NULL,
  verified_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  status VARCHAR(20) CHECK (status IN ('VERIFIED', 'WRONG_HALL', 'FLAGGED', 'ABSENT')),
  notes TEXT,
  sync_status BOOLEAN DEFAULT TRUE,
  UNIQUE(client_event_id)
);

CREATE INDEX idx_verification_mapping ON verification_logs(mapping_id);
CREATE INDEX idx_verification_status ON verification_logs(status);
```

`client_event_id` is generated on-device at scan time and used as an idempotency key so that repeated offline-sync retries never create duplicate log rows.

2.6 invigilators

```
CREATE TABLE invigilators (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,
  assigned_hall_id UUID REFERENCES halls(id),
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

2.7 admins

```
CREATE TABLE admins (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  password_hash TEXT NOT NULL,  
  role VARCHAR(20) CHECK (role IN ('SUPER_ADMIN', 'EXAM_STAFF', 'AUDITOR')) NOT NULL DEFAULT 'EXAM_STAFF',  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

3. Redis Data Structures

Key Pattern	Type	Purpose	TTL
session:{userId}	Hash	Active session/refresh-token metadata	Matches refresh-token lifetime
pass:render:{mappingId}	String	Current rotating display token for a student's barcode	60-120s
reveal:queue	Sorted Set	Scheduled reveal jobs, scored by unlock timestamp	N/A (persistent, consumed by worker)
dashboard:{examId}	Hash	Cached live counts (verified/pending/absent/wrong-hall) per hall	Refreshed on each event
ratelimit:{ip or userId}:{endpoint}	String (counter)	Login/verify endpoint rate limiting	Sliding window, seconds

4. Data Retention & Audit Notes

- `verification_logs` is treated as append-only at the application layer: no DELETE or UPDATE endpoint is exposed; corrections are made by inserting a new `FLAGGED` entry referencing the original via `notes` , preserving full history.
- `photo_url` values reference object storage keys, not public URLs; the API layer mints short-lived signed URLs on demand for both the student portal and the invigilator app's pre-sync step.
- Recommended retention: `verification_logs` and `student_exam_mappings` retained for a minimum of one academic year for audit/compliance purposes, with archival to cold storage thereafter per institutional policy.

5. Referential Integrity Summary

Relationship	Cardinality	On Delete
students → student_exam_mappings	1 : N	CASCADE
exams → student_exam_mappings	1 : N	CASCADE
halls → student_exam_mappings	1 : N	CASCADE
student_exam_mappings → verification_logs	1 : N	RESTRICT (preserve audit trail)
invigilators → verification_logs	1 : N	RESTRICT (preserve audit trail)